

aloud



Learn it by *saying it*.

Drop your notes. Play them out loud. In some levels the examiner lies to you, and you catch it with the page as proof.

You can recognise the answer on the page. Can you say it?

WHO IT IS FOR

A student revising from their own text notes before an oral exam, a viva or a seminar, who has read the pages several times and never tried to explain them aloud.

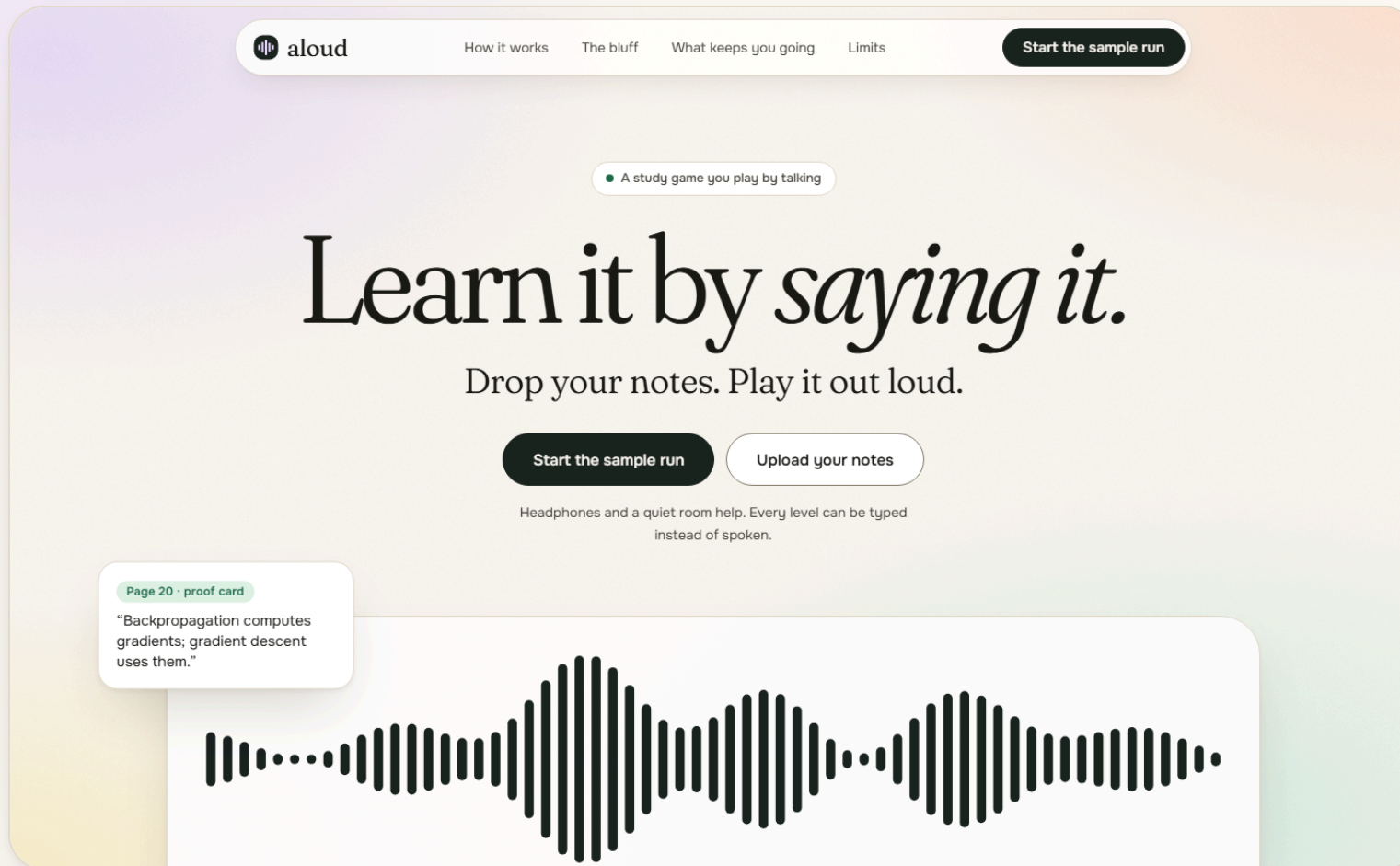
WHO IT IS NOT FOR

Scanned notes with no text layer, or anyone who wants their notes fact-checked. Aloud checks what you say against your notes, not the notes against the world.

Rereading and highlighting test recognition. Saying it with the page closed tests recall.

Alone, nobody asks the follow-up, and nothing pushes back when you nod along to a plausible sentence.

So Aloud pushes back. It states claims from your notes, half of them quietly altered, and the round is only won if you catch them.



Drop your notes. Play it out loud.

One click on the front page builds a run from the sample notes. Every level can also be typed, so a missing microphone does not stop the game.

WHAT THIS IS

Real screenshot of the front page, 1440 px, 2026-09-30.

THE SAMPLE RUN

22 levels in 5 worlds, built from Transformers notes written for this project.

PROOF CARD

The quote is accepted only if it is verbatim in the page it names.

Four kinds of level, one loop: play, score, come back.

Say it

Explain a concept. The examiner follows up. Code grades you against your pages.

Catch it

The examiner states a claim. Half are real, half altered. Call real or bluff.

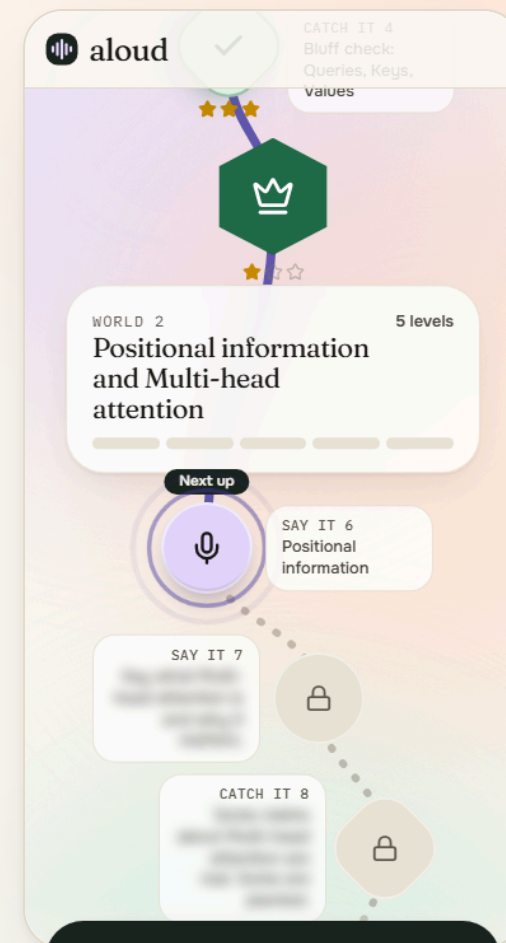
Boss

A mixed round that ends each world, with 4 hearts.

Recall

Concepts you missed come back later as a level.

3 hearts a level. XP with a combo multiplier. Up to 3 stars. Ranks 1 to 30. A daily streak with freezes earned every 7 days. Rules are pure functions in `src/lib/game`, unit-tested.



Run map at 390 px. Progress was seeded into local storage for the shot.

The examiner says something wrong. Do you notice?

THE EXAMINER SAYS

Each token is not projected into three vectors: a query, a key, and a value.

Bluff: a negation was added

YOUR PAGE, P. 7

Each token is projected into three vectors: a query (what this token is looking for), a key, and a value.

```
verify_claim(claim)
-> contradicted, page 7, quote checked as a verbatim substring
of that page
```

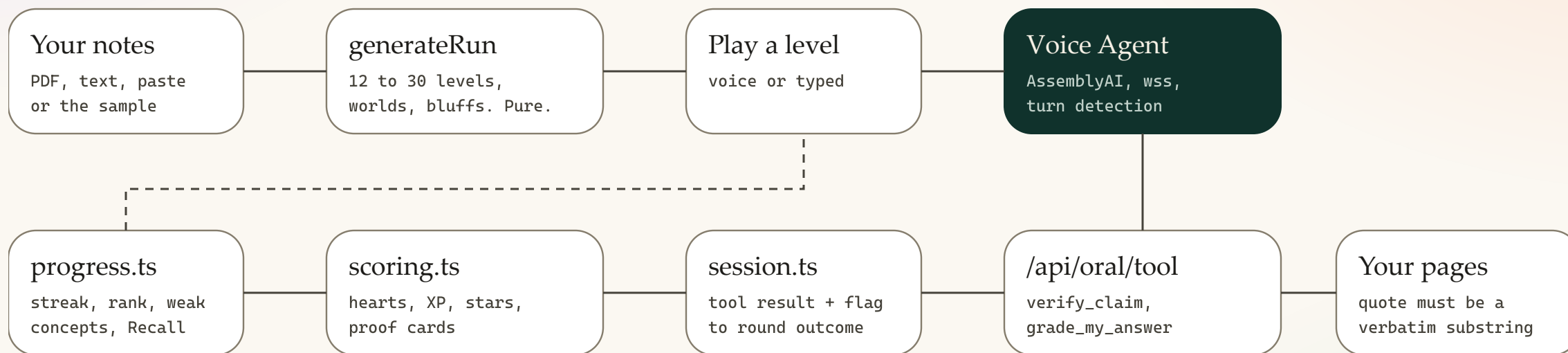
What decides the round

The stored flag on the claim and your call. The page check supplies the proof card. No code reads the examiner's prose, so a model that misspeaks cannot change your score.

Where it falls short

On the sample run the page check proved 7 of 11 bluffs. It missed 3 and called 1 supported. Real claims: 8 of 8 proved. A miss costs the proof line, not points.

Code builds the levels and keeps the score. The model only proposes a verdict.



Solid: one level. Dashed: progress feeds the next run map, and missed concepts return as Recall levels.

Typed play uses the same tools over HTTP. The account key stays on the server; the browser holds a token of at most 600 seconds.

The examiner is the Voice Agent API. Everything it says about your notes goes through a tool.

PIECE	WHAT ALOUD DOES
Voice Agent API	Browser WebSocket, 24 kHz PCM16 microphone audio up, examiner audio down.
Token flow	A server route mints a token of at most 600 s per connection. The key is in no client file; a test scans for it.
Session config	<code>session.update</code> with the versioned prompt, three tools, key terms from your concepts, and a block for the level being played.
Native turn detection	Only documented fields: <code>vad_threshold</code> , <code>interrupt_response</code> , <code>interruption_delay</code> . Undocumented names sent earlier were ignored by the service; probing found it.
Tool calling	<code>search_my_material</code> , <code>verify_claim</code> , <code>grade_my_answer</code> , held until <code>reply.done</code> , answered with <code>tool.result</code> .
Barge-in	On <code>input.speech.started</code> playback is flushed and late audio is dropped.

What was measured, how many times, and with what voice.

444_{ms}

Session ready. Median, n=3, 2026-09-29.

958_{ms}

First examiner audio. Median, n=3, 2026-09-29.

1286_{ms}

Learner's first loud sample to the service reporting speech. n=3. Not a measured playback stop.

54_{claims}

Labelled claims through verify_claim. 0 false supported. Judge median 590 ms.

7_{of 11}

Sample-run bluffs proven with a quote and page. 8 of 8 real claims proven. 2026-09-30.

22_{levels}

Sample run, built deterministically. All 26 subjects give 12 to 30.

1670_{tests}

Passed, 1 skipped, 100 files. 2026-09-30.

6_{screens}

0 axe violations at 390 px (wcag2a, wcag2aa, wcag21aa). Typed rounds, local.

Voice rows: synthetic learner voice (Windows System.Speech), a Node client running the browser's socket code, a local server. No browser and no human microphone. Numbers are in numbers.json.

Where it stops, said plainly.

- The game loop has not been played by voice with a human microphone in a browser. The take is the first run.
- The claim judge is a model. Code proves the quote is on the page, not that the verdict is right. 54 labelled claims is a small set.
- Typed Catch it rounds score from the flag and your tap. The correction field is not scored.
- Barge-in stop time in a browser was not measured. Reconnect starts a new session with recent turns.
- Chromium only. No phone, Firefox or Safari pass. Legal pages are drafts.
- Without a database the store is a temporary file. Progress is also kept in the browser.

LINKS

`github.com/vaibhav4046/aloud`

Live URL and video: on the submission form

BUILT ON

AssemblyAI Voice Agent API, Next.js, TypeScript

MADE BY

Vaibhav Lalwani, one person, September 2026